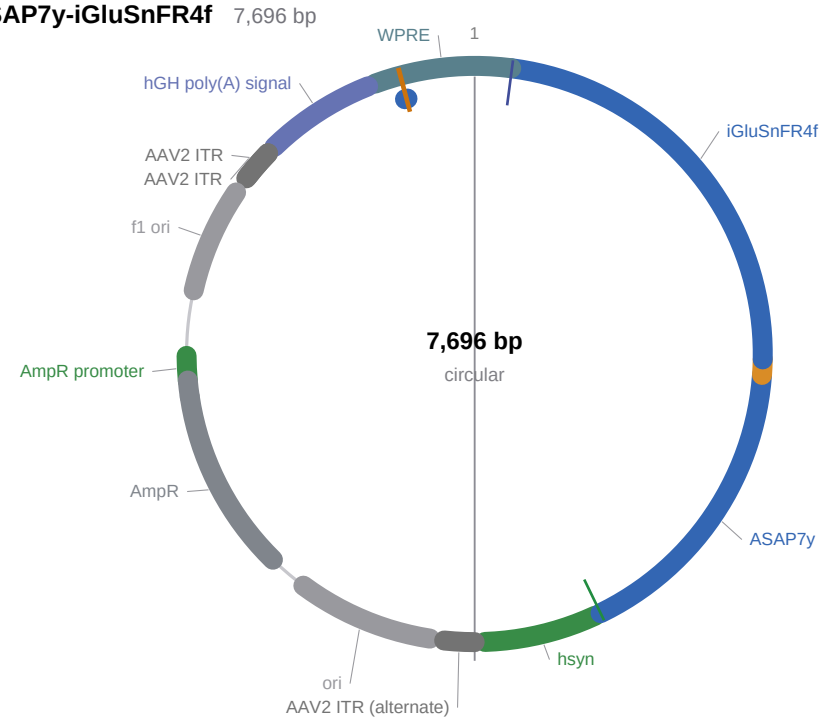
pAAV-hSyn-ASAP7y-iGluSnFR4f

PCR + NEBuilder HiFi assembly

Final construct	7,696 bp, circular
Backbone	pBJ027_pAAV_hSyn_ASAP3_ (5,829 bp, Bill Jia)
Architecture	hsyn - ASAP7y - P2A - iGluSnFR4f - WPRE - Factor Xa site - hGH poly(A) signal - sprBJ029 (HGH-pA REV) - AAV2 ITR - AAV2 ITR - f1 ori - AmpR promoter - AmpR - AmpR - ori - AAV2 ITR (alternate)
Workload	12 steps, 5 PCR(s), 1 assembly reaction(s)
Time	4.8 h hands-on, 2.8 days elapsed
Suggested name	pAAV-hSyn-ASAP7y-P2A-iGluSnFR4f

Designed construct

pAAV-hSyn-ASAP7y-iGluSnFR4f 7,696 bp



Outer ring reads forward, inner ring reverse. Position 1 is at the top.

- no stop codon at 4,163 in Factor Xa site
- no ATG (ATT) at 4,172 in Factor Xa site

Reagents and consumables

Quantities are per reaction. Check against the current package insert for the kits in use.

Item	Amount
5x high-fidelity reaction buffer	10 uL
10 mM dNTPs	1 uL
10 uM forward primer	2.5 uL
10 uM reverse primer	2.5 uL
template plasmid (1-10 ng)	1 uL
high-fidelity DNA polymerase	0.5 uL
nuclease-free water	to 50 uL
completed PCR reaction	all
DpnI	1 uL
agarose gel	0.8-1 % (1.5 % below 500 bp)
DNA stain	per kit

gel extraction kit	per kit
PCR or digest reaction	all
spin column or SPRI beads	per kit
2x assembly master mix	10 uL
linearised vector	50-100 ng
each insert	2:1 molar ratio over vector
competent cells	50 uL
assembly or ligation product	2 uL
SOC medium	950 uL
colonies	4-8
screening primer pair	per reaction
Taq master mix	per reaction
miniprep DNA	per facility spec
sequencing primers	one per junction

Primers to order

Lower case is the overlap or adapter tail; upper case binds the template. The quoted Tm is for the template-binding region only, which is what sets the annealing temperature.

Name	Dir	nt	Tm C	GC	Sequence (5' to 3')
backbone3-F	F	40	59.7	28%	ttgggcccctgttaaAATCAACCTCTGGATTACAAAATTT
backbone3-R	R	18	59.8	56%	AGGAACCCCTAGTGATGG
hsyn-F	F	78	61.3	58%	tcactaggggttcctgcggccgcacgcgtgtgtctagactgcagagggccctgcgtatgGTGTCTAGACTGCAGAGGG
hsyn-R	R	18	63.2	56%	TTCTCGACTGCGCTCTCA
ASAP7y-F	F	58	60.5	56%	gagcgcagtcgagaatcgagaaggtaccggatccgccaccATGGAGACGACTGTGAGG
ASAP7y-R	R	22	60.0	41%	GGTTACCACTTCAAGTTGTTTC
P2A-F	F	33	68.0	72%	cttgaagtggtaaccGGCTCCGGAGCCACGAAC
P2A-R	R	18	62.5	61%	GGGACCGGGGTTTTCTTC
iGluSnFR4f-F	F	34	60.8	53%	gaaaaccccggtcccATGGAGACAGACACACTCC
iGluSnFR4f-R	R	18	62.0	56%	TTAACAGGGGCCCAACAC

Protocol

1. PCR Amplify the 4,004 bp backbone3 arc (1,955-130) from pBJ027_pAAV_hSyn_ASAP3_ with primers backbone3-F / backbone3-R

annealing 60 C; extension 2:00 for 4,004 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

PCR reaction (50 uL, high-fidelity polymerase)

10 uL	5x high-fidelity reaction buffer
1 uL	10 mM dNTPs
2.5 uL	10 uM forward primer
2.5 uL	10 uM reverse primer
1 uL	template plasmid (1-10 ng)
0.5 uL	high-fidelity DNA polymerase
to 50 uL	nuclease-free water

2. PCR Amplify hsyn (485 bp) from HT75_LZF70_Fsyn_FAS_C extraction with primers hsyn-F / hsyn-R

annealing 61 C; extension 0:15 for 485 bp at 30 s/kb; 70% GC, add the high-GC enhancer; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

PCR reaction (50 uL, high-fidelity polymerase)

10 uL	5x high-fidelity reaction buffer
1 uL	10 mM dNTPs
2.5 uL	10 uM forward primer
2.5 uL	10 uM reverse primer

1 uL	template plasmid (1-10 ng)
0.5 uL	high-fidelity DNA polymerase
to 50 uL	nuclease-free water

3. PCR Amplify ASAP7y (1281 bp) from paav-ef1a-dio-asap7y-wpre_ with primers ASAP7y-F / ASAP7y-R

annealing 60 C; extension 0:38 for 1,281 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

PCR reaction (50 uL, high-fidelity polymerase)

10 uL	5x high-fidelity reaction buffer
1 uL	10 mM dNTPs
2.5 uL	10 uM forward primer
2.5 uL	10 uM reverse primer
1 uL	template plasmid (1-10 ng)
0.5 uL	high-fidelity DNA polymerase
to 50 uL	nuclease-free water

4. PCR Amplify P2A (66 bp) from pBJ230_pLJM1_JNKKTR_mK with primers P2A-F / P2A-R

annealing 62 C; primer Tm differ by 6 C, consider a gradient; extension 0:15 for 66 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

PCR reaction (50 uL, high-fidelity polymerase)

10 uL	5x high-fidelity reaction buffer
1 uL	10 mM dNTPs
2.5 uL	10 uM forward primer
2.5 uL	10 uM reverse primer
1 uL	template plasmid (1-10 ng)
0.5 uL	high-fidelity DNA polymerase
to 50 uL	nuclease-free water

5. PCR Amplify iGluSnFR4f (1791 bp) from YD043-pAAV-syn-iGluSnFR4f-NGR-WPRE with primers iGluSnFR4f-F / iGluSnFR4f-R

annealing 61 C; extension 0:54 for 1,791 bp at 30 s/kb; 98 C 30 s initial, 30 cycles, 72 C 2 min final.

PCR reaction (50 uL, high-fidelity polymerase)

10 uL	5x high-fidelity reaction buffer
1 uL	10 mM dNTPs
2.5 uL	10 uM forward primer
2.5 uL	10 uM reverse primer
1 uL	template plasmid (1-10 ng)
0.5 uL	high-fidelity DNA polymerase
to 50 uL	nuclease-free water

6. DPNI_DIGEST Digest the PCR reactions with DpnI to remove the parental plasmid templates (HT75_LZF70_Fsyn_FAS_C extraction, YD043-pAAV-syn-iGluSnFR4f-NGR-WPRE, pBJ027_pAAV_hSyn_ASAP3_, pBJ230_pLJM1_JNKKTR_mK)

37 C for 30-60 min, added directly to the PCR. Only cuts methylated (dam+) parental plasmid, not the amplicon.

7. GEL_EXTRACTION Gel purify hsyn (multiple products or a size-selection requirement)

Run far enough to resolve the bands, excise the correct band on a blue/long-wave transilluminator, elute in 20-30 uL.

8. DNA_CLEANUP Column purify the remaining PCR products

Elute in 20-30 uL. Quantify before assembly.

9. HI_FI_ASSEMBLY Assemble 5 fragments (hsyn + ASAP7y + P2A + iGluSnFR4f + backbone3) using NEBuilder HiFi, 25 bp overlaps

50 C for 15 min (2-3 fragments) or 60 min (4 or more). Keep total fragment input near 0.03-0.2 pmol.

HiFi / Gibson assembly (20 uL)

```
10 uL                  2x assembly master mix
50-100 ng              linearised vector
2:1 molar ratio over vector each insert
to 20 uL               nuclease-free water
```

10. `TRANSFORMATION` Transform the assembly into competent cells and plate on the appropriate antibiotic

Ice 30 min, 42 C for 30 s, ice 5 min, add SOC, recover 37 C for 1 h with shaking, plate on selective agar, incubate overnight at 37 C.

Transformation

```
50 uL                  competent cells
2 uL                   assembly or ligation product
950 uL                 SOC medium
```

11. `COLONY_SCREEN` Screen 4-8 colonies by colony PCR, or diagnostic digest with BamHI

Patch each colony before screening. Confirm candidates by diagnostic digest and set up overnight cultures.

12. `SEQUENCE_VERIFY` Sanger sequence across every junction, then confirm the full insert

Sequence across every junction first, then the full insert. Check the translated fusion, not only the DNA.

Detail

Everything behind the protocol above: how the design was validated, where each fragment comes from, what the other routes would cost, and the final sequence.

Validation

0 error(s), 6 warning(s), 5 informational.

Warnings

- dna: hsyn: 11 candidate parts matched; using 'hsyn'
- dna: P2A: 12 candidate parts matched; using 'P2A'
- dna: iGluSnFR4f: 'iGluSnFR4f' has Myc tag fused into it; the construct will contain that element even though it is not a separate module
- dna: the backbone feature 'Kozak sequence' straddles an insertion point and was disrupted
- protein: fusion (Factor Xa site) does not begin with ATG
- protein: fusion (Factor Xa site) does not end in a stop codon

Checks

- protein: predicted fusion protein (ASAP7y + P2A + iGluSnFR4f): 1045 aa
- protein: predicted fusion protein (Factor Xa site): 4 aa
- dna: unique sites available for diagnostic digest: AgeI, BglII, BsrGI, ClaI, EcoRV, MluI, SalI, XhoI
- packaging: carries the full AAV transfer plasmid element set
- verification: read back 4 of 4 parts from the assembled sequence, each once, in the specified order and orientation

Route warnings

- template HT75_LZF70_Fsyn_FAS_C extraction is not sequence-verified
- template paav-ef1a-dio-asap7y-wpre_ is not sequence-verified
- P2A is only 66 bp; rather than a separate PCR, add it to an adjacent fragment's primer tail or order it as annealed oligos
- template pBJ230_pLJM1_JNKKTR_mK is not sequence-verified
- template YD043-pAAV-syn-iGluSnFR4f-NGR-WPRE is not sequence-verified
- junction 10 (AAV2 ITR|f1 ori): overlap arm is 76% GC; HiFi joining may be poor
- junction 11 (f1 ori|AmpR promoter): overlap arm is only 16% GC; overlap Tm will be low
- junction 12 (AmpR promoter|AmpR): overlap arm is only 24% GC; overlap Tm will be low
- primer backbone3-F: GC content 28% is low
- primer backbone3-F: 6 nt self-complementarity at the 3' end; primer-dimer risk
- primer backbone3-R: annealing region occurs 2 times in the template; expect mispriming
- primer hsyn-F: GC-rich 3' end (4+ of the last 5 bases); may misprime
- primer ASAP7y-F: possible hairpin with a 6 bp stem
- primer P2A-F: annealing Tm 68.0 C is well above the 60 C target
- primer P2A-F: GC content 72% is high
- primer iGluSnFR4f-F: 4 nt self-complementarity at the 3' end; primer-dimer risk

Fragments and sources

Fragment	Length	Source template	Synthesis
hsyn	485 bp	HT75_LZF70_Fsyn_FAS_C extraction	no
ASAP7y	1,281 bp	paav-ef1a-dio-asap7y-wpre_	no
P2A	66 bp	pBJ230_pLJM1_JNKKTR_mK	no
iGluSnFR4f	1,791 bp	YD043-pAAV-syn-iGluSnFR4f-NGR-WPRE	no

Junctions

#	Left	Right	Frame shift	Notes
1	hsyn	ASAP7y	0	ok
2	ASAP7y	P2A	0	ok

3	P2A	iGluSnFR4f	0	ok
4	iGluSnFR4f	WPRE	0	ok
5	WPRE	Factor Xa site	0	ok
6	Factor Xa site	hGH poly(A) signal	0	ok
7	hGH poly(A) signal	sprBJ029 (HGH-pA REV)	0	ok
8	sprBJ029 (HGH-pA REV)	AAV2 ITR	0	ok
9	AAV2 ITR	AAV2 ITR	0	ok
10	AAV2 ITR	f1 ori	0	overlap arm is 76% GC; HiFi joining may be poor
11	f1 ori	AmpR promoter	0	overlap arm is only 16% GC; overlap Tm will be low
12	AmpR promoter	AmpR	0	overlap arm is only 24% GC; overlap Tm will be low
13	AmpR	AmpR	0	ok
14	AmpR	ori	0	ok
15	ori	AAV2 ITR (alternate)	0	ok

Route comparison

Route	Steps	PCR	Assembly	Hands-on	Elapsed	Cost	Usable
PCR + NEBuilder HiFi assembly (this one)	12	5	1	4.8 h	2.8 d	108	yes
Swap by digest + one PCR + HiFi assembly	0	0	0	0.0 h	0.0 d	1000	no
Swap by PCR + HiFi assembly (2 parents)	0	0	0	0.0 h	0.0 d	1000	no
Swap by restriction digest + ligation (2 parents)	0	0	0	0.0 h	0.0 d	1000	no
Restriction digest + ligation	0	0	0	0.0 h	0.0 d	1000	no

Swap by digest + one PCR + HiFi assembly was not usable: 2 of the parts this design needs are absent from pBJ027_pAAV_hSyn_ASAP3_ (P2A, iGluSnFR4f), and a single swap replaces a single region, so no edit of this plasmid produces the construct asked for; build it from parts, or start from a recipient that already carries the rest

Swap by PCR + HiFi assembly (2 parents) was not usable: 2 of the parts this design needs are absent from pBJ027_pAAV_hSyn_ASAP3_ (P2A, iGluSnFR4f), and a single swap replaces a single region, so no edit of this plasmid produces the construct asked for; build it from parts, or start from a recipient that already carries the rest

Swap by restriction digest + ligation (2 parents) was not usable: 2 of the parts this design needs are absent from pBJ027_pAAV_hSyn_ASAP3_ (P2A, iGluSnFR4f), and a single swap replaces a single region, so no edit of this plasmid produces the construct asked for; build it from parts, or start from a recipient that already carries the rest

Restriction digest + ligation was not usable: the insert is made of 4 pieces from 4 donor plasmid(s); a ligation of that many fragments is not practical, so this construct needs the PCR/HiFi route (or a staged build of two pieces at a time)

Final sequence

7,696 bp, 53.4% GC, circular.

```
1  gcggccgcac gcgtgtgtct agactgcaga gggccctgcg tatggtgtct agactgcaga
61  gggccctgcg tatgagtgca agtgggtttt aggaccagga tgaggcgggg tgggggtgcc
121  tacctgacga ccgaccccga cccactggac aagcacccaa cccccattcc ccaaattgcg
181  catcccctat cagagagggg gaggggaaac aggatgcggc gaggcgcgtg cgcactgcca
241  gcttcagcac cgcggacagt gccttcgccc ccgcctggcg gcgcgcgcca ccgccgcctc
301  agcactgaag gcgcgctgac gtcactcgcc ggtcccccgc aaactcccct tcccggccac
361  cttggtcgcg tccgcgccgc cgccggccca gccggaccgc accacgcgag gcgcgagata
421  gggggcacg ggcgcgacca tctgcgctgc ggcgccggcg actcagcgct gcctcagtct
481  gcggtgggca gcggaggagt cgtgtcgtgc ctgagagcgc agtcgagaat cgagaaggta
541  ccggatccgc caccatggag acgactgtga ggtatgaaca ggggtcagag ctcactaaaa
601  cttcgagctc tccaacagca gatgagccca cgataaagat tgatgatggt cgtgatgagg
661  gtaatgaaca agacagctgt tccaatacca ttaggagaaa aatttccccg tttgtgatgt
721  catttggatt cagagtattt ggagttgtgc ttatcattgt agacatcata gtggtgattg
781  tggatctggc catcagtgag aagaaaagag gcattagaga gattcttgaa ggtgtttccc
841  tggctatagc actcttcttc cttgttgatg ttctcatgag agtgtttgtt gaaggcttca
901  agaactattt ccggtccaaa ctgaatactt tggatgcagt catagtagtg ggcactctgc
961  taattgctat gacctactcc ttctctgctc ttgctgcctt taaatccgag aacgtgtata
1021  ttaccgcgga taaacagaaa aacggcatta aagcgaactt taccgtgatt cataacgtgg
1081  aagatggcag cgtgcagctg gcggatcatt atcagcagaa caccccgatt ggcgatggcc
1141  cggtgctgct gccggataac cattatctga gctaccagac cgtgctgagc aaagatccga
1201  acgaaaaacg cgatcacatg gtgctgctgg aagacgtgac cgcagcgggc attacacacg
```

```
1261  gcatggatga actgtatggc ggcaccggcg gcagcgcgag ccagggcgaa gaactgttta
1321  ccggcgtggt gccgattctg gtggaactgg atggcgatgt gaacggccat aaatttagcg
1381  tgcgcggcga aggcgaaggc gatgcgacca ttggcaaact gaccctgaaa tttatttgca
1441  ccaccggcaa actaccggtg ccgtggccga ccctggtgac caccttaacc tatggcgtgc
1501  agtgctttag ccgctatccg gatcatatga aacgccatga ttttttttaaa agcgcgatgc
1561  cggaaggcta tgtgcaggaa cgcaccatta gctttaaaga tgatggcaaa tataaaaccc
1621  gcgcggtggt gaaatttgaa ggcgataccc tggtgaaccg cattgaactg aaaggcaccg
1681  attttaaaga agatggcaac attctggggc ataaactgga atataacatt gatcagatgc
1741  cgcagatggt tactcttctt cgagttctga gaattgttat cttaataaga gtgattcgcc
1801  tggcttcaca gaagaaacaa cttgaagtgg taaccggctc cggagccacg aacttctctc
1861  tgttaaagca agcaggagac gtggaagaaa accccggtcc catggagaca gacacactcc
1921  tgctatgggt actgctgctc tgggttccag gttccactgg tgacggatcc gccgcaggca
1981  gcacgctgga caaaatcgcc aaaaacggtg tgattgtcgt cggtcaccgt gactcttcag
2041  tgcctttctc ttatcaagac agtcagcaca aagtggtggg ttactcgcag gaatactcca
2101  acgccattgt tgaagcagtg aaaaagaaac tcaacaaacc ggacttgcag gtaaaactga
2161  ttccgattac ttcacaaaac cgtattccac tactgcaaaa cggcacttac gattttgaat
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2281  tcgtggtcgg tacgcgcctg ttgaccaaaa agggtggcga tatcaaagat tttgccaacc
2341  tgaaagacaa agccgtagtc gtcacttccg gcactacctc tgaagttttg ctcaacaaac
2401  tgaatgaaga gcaaaaaatg aatatgcgca tcatcagcgc caaagatcac ggtgactctt
2461  tccgcaccct ggaaagcggt cgtgccgttg cctttatgat ggatgacgta ctgctggccg
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2701  aaaatccaat tctggtgaac cacacggtct atatcaccgc cgacaagcag aagaacggca
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3121  ccaccaacgg caagctaacc ctgaagttca tctgcaccac cggcaagctg cccgtgccct
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3241  acatgaagca gcacgacttc ttcaagtccg ccatgcccga aggctacgtc caggagcgca
3301  ccatcagctt caaggacgac ggaacctaca agacccgcgc cgaggtgaag ttagagggcg
3361  acaccctggt gaaccgcatc gagctaaagg gcatcgactt caaggaggac ggcaacatcc
3421  tggggcacaa gctggagtac aactttaaca accccctggt catgaatttc gaactgtcag
3481  acgaaatgaa agcactgttc aaagaaccga atgacaaggc actgaaactg caggtcgacg
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3601  cagaaggatc aggtgccctt cctgcccttg catgtagtct ggctccactg ggactggcat
3661  tggttctgtg gacggtgttg ggcccctgtt aaaatcaacc tctggattac aaaatttgtg
3721  aaagattgac tggtattctt aactatgttg ctccttttac gctatgtgga tacgctgctt
3781  taatgccttt gtatcatgct attgcttccc gtatggcttt cattttctcc tccttgtata
3841  aatcctggtt gctgtctctt tatgaggagt tgtggcccgt tgtcaggcaa cgtggcgtgg
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4441  tattatgggg tggaggggag tggtatggag caaggggcaa gttgggaaga caacctgtag
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4561  aatctccgcc tcctgggttc aagcgattct cctgcctcag cctcccgagt tgttgggatt
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4801  ggtaaccacg tgcggaccga gcggccgcag gaacccctag tgatggagtt ggccactccc
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4981  cggtattttc tccttacgca tctgtgcggt atttcacacc gcatacgtca aagcaaccat
5041  agtacgcgcc ctgtagcggc gcattaagcg cggcgggtgt ggtggttacg cgcagcgtga
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5641  gctgcatgtg tcagaggttt tcaccgtcat caccgaaacg cgcgagacga aagggcctcg
5701  tgatacgcct atttttatag gttaatgtca tgataataat ggtttcttag acgtcaggtg
```

```
5761  gcacttttcg gggaaatgtg cgcggaaccc ctatttgttt atttttctaa atacattcaa
5821  atatgtatcc gctcatgaga caataaccct gataaatgct tcaataatat tgaaaaagga
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